

HELIOLITE · REQUIREMENTS

HelioLite requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/heliolite		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	R15 budget raised to \$430 (HLT-DDR-001 D4); R5 allows the decided anemometer (D2); status column replaced by the TRL 3 results of HLT-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

HelioLite requirements

These are the requirements for the concept, checked by calculation at TRL 3 in HLT-CAL-001. Targets are proposals, not yet validated with users. "Met" means met on paper only. On 2026-09-25 Amish decided to raise the budget to \$430 (HLT-DDR-001, D4), so R15 now reads \$430, and R5 now names the weather anemometer decided in D2 as allowed; no other target has changed. Proposed rewordings of R7 and R10 are awaiting Amish and are listed below.

The **design point** used throughout is a clear winter day at 45° N, DNI 800 W/m², a mirror-to-target distance of 10 m and an angle of incidence on the mirror of about 32° (cosine factor 0.85).

Table 1. Requirements and TRL 3 status (tags refer to lines printed by docs/04-calcs/sizing.py).

ID	Requirement	Target	Status (HLT-CAL-001)	Verification
R1	Redirected sunlight at the target	200 W or more arriving at the outside of the target glazing at the design point	Met: 211 W [A2]	Optical calculation (done)
R2	Useful daylight in the room	Raise the mean illuminance of a 12 m ² room by 300 lx or more at the design point	Met: 502 lx, utilization 0.4 assumed [A3]	Illuminance calculation (done)
R3	Daily energy delivered	0.5 kWh or more through the glazing on a clear winter day	At risk: 0.50 kWh on 21 Dec and 1.00 kWh on 1 Feb at reference site A; 0.40 kWh at site B on 21 Dec [B2, B3]	Hourly sun-path calculation (done); siting survey
R4	Pointing accuracy	Beam center within 0.5° of the target center (about 90 mm at 10 m), with winds up to 8 m/s	At risk: 0.31° typical and 0.50° at the 95th percentile, only with four-point calibration and preloaded drives; 1.04° without preload [D5, D6]	Error budget and calibration simulation (done); later spot log
R5	Sensorless tracking	Sun position from time and location only; no sun sensor or camera; axis homing switches and a weather anemometer allowed	Met by design	Design review
R6	Timekeeping	Clock error 60 s or less after 6 months without a network time source	Met: 31.5 s (DS3231, ± 2 ppm) [D1]	Datasheet check
R7	Calibration	One person sets up and calibrates from a phone in 30 min or less, with no survey instrument	At risk: 19 min hands-on for four points, but the points span about 4 h [G1, G2]	Task analysis (done)
R8	Working range	Target distance 3 to 20 m; mirror normal azimuth $\pm 135^\circ$ about the target direction; elevation of the normal -90° (face-down stow) to $+90^\circ$	Met mechanically; spot plus 0.5° error fits a 1.0 m window up to 16.8 m [A6, K1]	Geometry check (done)
R9	Wind	Hold R4 up to 8 m/s; survive 35 m/s (126 km/h) gusts in stow without damage	Not met: assumed stowed hinge moment 24.3 N·m at 35 m/s against 17 N·m rated and 22 N·m listed maximum for the NMRV030-class gearbox [E2, E3]	Wind load calculation (done); gearbox data
R10	Safe beam when stowed, faulted or unpowered	Beam directed only at the target or the ground within 3 m of the mast; stow face-down within 60 s of a fault or power loss	At risk: stow in 23 s on the supercapacitor reserve with 2.1x energy margin, beam moving only downward, but crossing the ground beyond 3 m for about 2.8 s [F1, F2, F4]	Calculation (done); later fault test
R11	Electrical safety	Only 12 V DC (SELV) outdoors; electronics enclosure IP65; mains adapter indoors and listed	Met by design	Design review
R12	Outdoor life	Operate from -20 to $+45$ °C; UV-stable plastics (ASA or better);	Not verifiable at TRL 3	Materials review; later exposure test

ID	Requirement	Target	Status (HLT-CAL-001)	Verification
		corrosion-resistant mast; 10-year mirror life with cleaning		
R13	Mass and install	10 kg or less on the mast top; installed by two people with hand tools in 4 h or less	Not met: 12.5 kg on the mast top [H3]; install time not verifiable at TRL 3	Mass estimate (done)
R14	Standby power	3 W or less average from the 12 V supply	Met: 0.60 W [I1]	Power budget (done)
R15	Cost	Parts \$430 or less (budget raised from \$400, decided by Amish 2026-09-25)	Met, thin: \$425 [J1]	Priced BOM (bom/ bom.csv)

■ Requirements not met or at risk

- **R9 (wind survival) not met.** Storm awareness (anemometer and forecast, decided) makes the unit stow at a 15 m/s gust, where the hinge moment is 7.4 N·m. In stow, the assumed stowed hinge moment at 35 m/s exceeds the gearbox's listed maximum. Options are proposed in HLT-CAL-001, section E.
- **R13 (mass) not met.** Rated NMRV030-class gearboxes and stiffer yoke arms bring the mast-top mass to 12.5 kg.
- **R3 (daily energy) at risk.** Site dependent: met at site A with no margin on the solstice, missed at site B.
- **R4 (pointing) at risk.** Relies on four-point calibration over about 4 h and on a preload on each drive (proposed, awaiting Amish) against the gearbox's 1° backlash.
- **R7 (calibration) at risk.** Hands-on time is within target, elapsed time is not. Proposed rewording, awaiting Amish: "in 30 min or less of hands-on time, spread over one clear day".
- **R10 (safe beam) at risk.** The stow itself works on stored energy. Proposed clarification, awaiting Amish: "during a stow the beam may move only downward, toward the ground".
- **R12 (outdoor life) not verifiable at TRL 3.**
- **R15 (cost)** is met with \$5 of margin; preload springs (about \$6) would use it up.

■ Assumptions

- Mirror area 0.36 m² (600 x 600 mm), solar-weighted reflectance about 0.93 when clean and 0.95 soiling factor (combined 0.88).
- Double glazing transmits about 75 % of the beam at normal incidence, less at oblique incidence (HLT-CAL-001).
- Luminous efficacy of direct sunlight about 95 lm/W; a utilization factor of 0.4 for sunlight bounced from a light ceiling or a diffuser into a 3 x 4 m room.
- R3 is checked by an hourly clear-sky model at three illustrative sites at 45° N (HLT-CAL-001, section B). The TRL 2 shortcut of six useful hours at a mean DNI of 650 W/m² and a mean cosine factor of 0.80 gives 0.73 kWh and is no longer used.